

ONC-305-301

Programming and QC Plan

Reproducibility, validation and programming standards

Document	Programming and QC Plan	Study	ONC-305-301
Version	1.0	Status	Final simulated package
Data cut-off	2026-01-31	Database lock	2026-02-15
Author	Luke Johnston	Classification	Portfolio demonstration
Purpose	Biostatistics/statistical-programming portfolio	Data	100% simulated; no patient data

Important: this is a simulated portfolio package designed to demonstrate biostatistics, statistical programming, reproducibility, and regulatory-document writing. It is not a real clinical trial and is not clinical evidence.

1. Programming model

- The portfolio follows a clinical-trial programming flow: source data, ADaM-style derivation, TLF generation, validation checks, and report rendering.
- Each output has a program owner and traceability to source datasets and methods.
- The repository is designed so a recruiter can run one command and regenerate the package.

2. Validation strategy

Area	Primary checks	Evidence
Population counts	ADSL counts by arm and flags; FAS/SAF consistency	validation_checks.csv
Endpoint integrity	PFS/OS record count, CNSR values, positive AVAL, event/censor definitions	ADTTE + TLF checks
Safety chronology	AE start/end dates, treatment-emergent flags, toxicity-grade range	ADAE checks
Output production	Expected tables, listings, figures and reports exist	Output existence checks
Metadata	ADaM specification and define-like XML are generated	data/specs
Visual review	PDFs rendered and inspected for clipping/overlap	Rendered preview assets

3. Program validation tracker

Program	Purpose	Role	Validation approach	Status
generate_data.py	Synthetic SDTM-like source data	Primary	Code review + raw_manifest counts	PASS
build_adam.py	ADaM-style datasets and metadata	Primary	Traceability review + variable-level checks	PASS
generate_tlfs.py	Tables/listings/figures	Primary	Output existence + selected hand calculations	PASS
qc_validation.py	Automated validation checks	Primary	PASS/WARN/FAIL tracker	PASS
render_reports.py	Baseline report rendering	Primary	PDF render verification	PASS
render_professional_package.py	Professional SAP/CSR-style package	Primary	PDF render verification + artifact manifest	PASS
programs/sas/*.sas	SAS templates	Template	Provided as regulated-workflow examples; not executed in Python environment	TEMPLATE
programs/r/*.R	R templates	Template	Provided as alternate implementation examples	TEMPLATE

4. Automated validation check summary

Current run: 62 PASS, 0 WARN, 0 FAIL across 62 checks.

Check	Status	Detail
ADSL subject uniqueness	PASS	ADSL subjects=420, unique=420
ADSL equals raw DM count	PASS	ADSL=420, DM=420
Treatment arms present	PASS	Arms=['ONC-305 + SOC', 'Placebo + SOC']
Planned and actual treatment match in simulation	PASS	All actual treatments equal planned treatment
FAS flag complete	PASS	FAS N=420
Safety flag complete	PASS	Safety N=420
ITT flag complete	PASS	ITT N=420
Randomization before or on treatment start	PASS	RANDDT <= TRTSDT for all subjects
Treatment start before or on treatment end	PASS	TRTSDT <= TRTEDT for all subjects
Positive treatment duration	PASS	Minimum TRTDURD=26
Randomization roughly balanced	PASS	ONC-305 + SOC=211, Placebo + SOC=209
ADTTE has two records per subject	PASS	ADTTE rows=840, expected=840
ADTTE unique subject/endpoint	PASS	No duplicate USUBJID/PARAMCD records
ADTTE endpoints are PFS and OS	PASS	PARAMCD=['OS', 'PFS']
ADTTE censoring values valid	PASS	CNSR values=[0, 1]
ADTTE event description consistent	PASS	EVNTDESC=['Censored', 'Event']
ADTTE event/censor consistency	PASS	CNSR and EVNTDESC agree
ADTTE positive times	PASS	Minimum AVAL=0.05
ADTTE dates after randomization	PASS	ADT >= STARTDT for all records
ADTTE analysis flag complete	PASS	ANL01FL Y count=840
ADRS one record per subject	PASS	ADRS subjects=420
ADRS BOR values valid	PASS	BOR=['CR', 'NE', 'PD', 'PR', 'SD']
ADRS ORR flag consistent	PASS	ORRFL equals Y for CR/PR only
ADRS DCR flag consistent	PASS	DCRFL equals Y for CR/PR/SD only
ADLB four parameters per subject	PASS	ADLB rows=1680, expected=1680
ADLB unique subject/parameter	PASS	No duplicate lab records by subject/parameter
ADLB grades valid	PASS	Grades=[np.int64(0), np.int64(1), np.int64(2), np.int64(3), np.int64(4)]
ADAE row count matches raw AE	PASS	ADAE=976, raw AE=976
ADAE unique subject/sequence	PASS	No duplicate USUBJID/AESEQ records
ADAE treatment-emergent dates	PASS	All AE start dates are on or after treatment start
ADAE end dates after start dates	PASS	AEENDT >= AESTDT
ADAE toxicity grades valid	PASS	Grades=[1, 2, 3, 4, 5]
ADAE serious flag valid	PASS	AESER=['N', 'Y']
ADAE relatedness flag valid	PASS	AEREL=['N', 'Y']
ADAE fatal flag valid	PASS	AESDTH=['N', 'Y']

Check	Status	Detail
ADAE occurrence flags valid	PASS	AOCCFL values=['Y']
ADaM specification exists	PASS	adam_spec.csv present=True
Define-like metadata exists	PASS	define_like_metadata.xml present=True
ADaM spec covers five datasets	PASS	Datasets=['ADAE', 'ADLB', 'ADRS', 'ADSL', 'ADTTE']
Output table exists: t14_1_1_demographics	PASS	t14_1_1_demographics.csv/md/txt present=True
... truncated in PDF; full file included in repository		

5. How this mirrors regulated work

- The primary analysis is not just a model output; it is tied to a SAP, analysis population, endpoint definition, TLF, source dataset and QC check.
- The project distinguishes executable implementation from SAS/R templates, avoiding overstating unexecuted code.
- Reviewer-facing artifacts make the repository easier to inspect in an interview and closer to what pharma teams expect from analysis packages.